Nguyễn Thượng Quyền

Fullstack Developer & DevOps Engineer

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EDUCATION

Samsung Innovation Campus 2023.10 - 2024.12

Big Data

FPT University Hanoi 2022.09 - 2026.04

Software Engineering

HUS High School for Gifted Students 2019.09 - 2022.05

Chemistry

WORK EXPERIENCE

POSitive 2025.04 - 2026.03

Founder & Lead Developer

Founded and led the development of POSitive, a Point of Sale (POS) solution backed by FPT University to empower small and medium-sized enterprises (SMEs) in Vietnam. I architected a comprehensive platform to digitize and streamline business operations, successfully tackling key market challenges including manual inventory management, slow payment processing, and complex e-invoicing compliance: Demo

- Architected a dual-component system with a responsive Vue.js frontend for the sales interface and a robust Laravel backend for the admin portal.
- Engineered a high-speed payment module supporting multiple methods (QR Code, cash) and automated the generation of legally compliant electronic invoices.
- Reduced hardware costs for businesses by implementing a barcode scanning feature that utilizes the device's built-in camera.
- Developed real-time inventory and product management systems to provide owners with instant operational oversight.
- Built an intuitive admin dashboard featuring visual revenue reports, customer relationship management (CRM), and staff access controls.

TDA Global 2025.03 - 2025.09

Junior Fullstack Developer

- Develop business tools for internal company use:
    - End-to-end system for POD product image rendering: Mockup Automation
    - Paypal management tools
    - Internal Workspace for TDA Global: Set up and managed a company-wide workspace on a Network-Attached Storage (NAS) system, handling centralized file storage, internal email, and other services: TDA Cloud
    - Developed and maintained AI systems using Retrieval-Augmented Generation (RAG).
    - Built and updated AI workflows leveraging ComfyUI for model orchestration and experimentation.
- Develop Web and Mobile apps:
    - NFCite: A website that lets businesses create interactive web pages linked to NFC tags and QR codes using a visual drag-and-drop builder.
    - Biểu đồ giá vàng: A mobile app providing real-time gold price data.

MegaAds Software Technology and Online Advertising Joint Stock Company 2024.08 - 2025.01

Fresher Fullstack Developer

- Maintain, update source code of several E-commerce or POD platforms:
    - printerval.com (83.2K visits)
    - chiaki.vn (1.6M visits)
    - nhathuocsuckhoe.com (7.8K visits)
    - vuahanghieu.com (858.5K visits)
    - thegioigiay.com
    - customyourprint.com
- Develop development tools for internal company use
- Mainly use Laravel, ExpressJS and AngularJS

KIS Vietnam Creative Co., LTD

2024.02 - 2024.08

Collaborator Unity Developer

Develop 2D casual games using Unity engine:

- Seafood Merge (1K downloads)

Chillin' Studio

2023.05 - 2024.01

Unity Developer & Co-founder

Develop 2D casual games using Unity engine:

- Wonder World

Vietnam Robotics Challenge - FPT University

2023.05 - 2023.09

Technical Mentor

- Mentoring high school students in understanding the concept of Arduino-based machines
- Provide technical support in mechanical designing& programming using C++/Tinker Cad

HONORS & AWARDS

Runner-up (Solo work) - JS Club Coding Inspiration	2026
First Author (Approved) - IEEE AICSP	2026
Consolation Prize - JS Club Codecipher	2025
Runner-up - JS Club Algorithm Arena 6.0	2025
Consolation Prize - FPT Education Research Festival	2024
Consolation Prize - JS Club Coding Inspiration	2023

PROJECTS

Automated Grading Support System for Software Testing Course (SWT301) Version 2

2025.12 - 2026.05

Team Leader & Lead Fullstack Developer & DevOps

FPT University Capstone (SEP490)

Project description: Built an AI-powered platform to automate grading for Software Testing exams, featuring dynamic exam generation, automated grading guides, and a multi-agent grading system with human-in-the-loop review. Managed a project scope of 455 man-days across four iterative phases. Deployed on a VPS with a custom Telegram DevOps bot for remote management. Now being piloted by FPT University's Department of Software Engineering.

Team size: 4

Key Achievements:

- Achieved an outstanding score of 9.5 for the final capstone project.
- Commended by the defense committee for rigorous execution, clear task delegation, and delivering a high-quality Software Engineering thesis that thoroughly met all technical, content, and formatting requirements, successfully navigating a reduction in team size prior to the

defense.
- First author of an approved scientific paper at the 2026 IEEE Xplore AICSP conference.

Responsibilities:
- Developed a multi-modal indexing workflow utilizing Gemini Vision OCR for scanned files and direct parsing for Markdown and bookmarked PDFs. Implemented a structured three-layer chunking strategy that meticulously preserves sentence boundaries.
- Solved a critical retrieval accuracy problem by embedding header breadcrumbs alongside the first 2,000 characters into a Qdrant database, while simultaneously retrieving up to 5,000 characters as a payload to ensure the LLM receives complete context. Engineered a zero-cost synchronization feature where grading guides read context directly from the database, effectively eliminating redundant vector searches.
- Prevented database connection pool exhaustion during extended AI generation tasks by executing AI calls entirely independent of active database connections. Engineered distributed batch grading with Java 21 Virtual Threads, per-workflow semaphores, and Redis BLPOP queues to guarantee reliable, loss-free message delivery.
- Implemented a robust grading architecture where a Grader agent evaluates submissions, followed immediately by a fine-tuned GPT-4.1-mini Auditor agent in a verification loop. Configured confidence-based auto-approval for scores of 80 percent or higher, achieving an impressive 82 percent auto-approval rate in production.
- Built a strict dual-layer plagiarism system utilizing semantic HNSW search and lexical Jaccard similarity to confirm exact wording overlaps. This dual approach ensures zero false positives on legitimately correct student answers.
- Developed highly secure parsing for DOCX, XLSX, PDF, and ZIP files with built-in detection for merged cells, prompt injection attempts, encrypted files, and non-textual content overlays.
- Authored the continuous integration pipeline and a custom Telegram DevOps Bot to enable password-protected remote deployments, supervisor process control, live log streaming, and server monitoring.

Tech: Java 21, Spring Boot 3.5, React 19, TypeScript, PostgreSQL, Qdrant, Redis, AWS S3, Gemini API, OpenAI API, TailwindCSS, Bash, Telegram Bot API.
Link: https://agss.positive.com.vn/

Nguyệt

2026.06 - 2026.06

Team Leader & Sole Developer

Runner-up (2nd Place) - Coding Inspiration 2026

Project description: AI-companion cycle-health app that unifies period tracking, journaling, and task scheduling on one timeline, with a Gemini-powered AI agent that reasons over the user's own data to give personalized, cycle-aware guidance. Took the product from concept to a fully working, pilot-tested app in under two weeks.
Team size: 2
Key Achievements:
- **Runner-up (2nd Place)**, FPT Coding Inspiration 2026.
- Went from idea to a working app, piloted with real users at FPT University, in under two weeks.

Responsibilities:
- Led the team and owned the end-to-end product: backend, AI agent, and Android app.
- Designed the core AI agent with 32 tools so it always checks real user data before giving advice, instead of guessing.
- Built a smart cycle-prediction engine that adapts to each user instead of assuming a fixed 28-day cycle.
- Kept the AI's medical answers grounded in a curated knowledge base, refusing to answer rather than risk giving wrong health advice.
- Hardened the app for real users: encrypted health data, strict per-user privacy, and full account/data deletion on request.

Tech: Laravel 13, Vertex AI (Gemini 2.5), MySQL, Redis, Sanctum, React 19, Capacitor, FastAPI, NotebookLM, Firebase, Google Calendar API.
Link: https://luna.montserrat.id.vn/

ToySharing

2025.11 - 2025.11

Solo developer

Project description: A simple toy-sharing system, final project for PRN232 course at FPT University
Technologies: ASP .NET, AngularJS
Responsibilities: System design, full-stack development
Github: makecolour/FinalProject

EchoWave

2024.11 - 2024.11

Solo developer

Description: A production-ready messaging platform designed for horizontal scalability using PM2 clustering.

Key Engineering: Implemented Redis Pub/Sub to handle message broadcasting across multiple server clusters, ensuring reliable delivery in high-traffic scenarios.
Features: Built a companion Android app and a web interface (AngularJS) supporting offline message persistence and automatic device management.
Tech: Node.js, Express, Socket.IO, Redis, MySQL, AngularJS.
Github: makecolour/EchoWave

Voyagenius

2025.10 - 2025.11

Backend developer, system designer

National Finalist, NAVER Vietnam AI Hackathon
Role: Sole backend developer orchestrating the system for the National Finals.
Architecture: Designed a service mesh integrating 8+ NAVER Cloud services (Maps, Clova OCR, Papago, Green-Eye) with Amadeus GDS for real-time travel booking.
AI Agent: Engineered a context-aware AI workflow allowing users to plan trips via natural language, utilizing RAG systems to fetch live travel data.
Team size: 4
Tech: Laravel 12, Sanctum, Redis, OpenAPI, Docker.
Link: https://travel-naver-client-ofpw.vercel.app

Smart To-Do List

2025.09 - 2025.09

Solo developer

NAVER Vietnam AI Hackathon – Preliminary Round Qualifier
Description: An intelligent, procrastination-aware task manager that qualified the team for the National Finals.
AI Features: Integrated OpenAI GPT-4 to analyze user habits, assigning a "procrastination score" (0.0-9.9) and suggesting behavioral interventions.
Real-time: Utilized Pusher and Redis for instant cross-device synchronization and Pomodoro session tracking.
Tech: React 19, Laravel, TailwindCSS, OpenAI API.
Link: https://web-track-naver-vietnam-ai-hackatho-ecru.vercel.app

Clowndinary

2025.08 - 2025.09

Solo developer

Project description: A web application to simplify media uploads to the multiple CDN platforms through a clean, user-friendly interface.
Customers: TDA Global
Technologies: ExpressJS
Responsibilities: System design, full-stack development, Cloudinary and Bunny Storage API integration.
Github: makecolour/Clowndinary

Trust Management System

2024.05 - 2024.08

Researcher, fullstack developer

Project description: Developed based on the research document "A Novel Approach of Reputation Trust Metric Based on Knowledge Graph Combined with Large Language Model for Improving Trust Worthiness Calculation on Social Networks" to visualize and provide results from the research process. Consists of a web application and a Chromium extension for Facebook.
Team size: 3
Technologies: Maven, Java SpringBoot, ChromeAPIs
Github: makecolour/TrustScore - makecolour/TrustScoreExtension
Develop research tools and Trust Management System to bring our research to industry environment

SKILLS

DevOps & Infra	Docker, CI/CD (GitHub Actions), Bash scripting, VPS Deployment, PM2 Clustering, AWS S3, Redis Pub/Sub, Telegram Bot API
Programming languages	PHP, JavaScript/TypeScript, Java, C#, HTML/CSS,...
Frameworks	Laravel, .NET, SpringBoot, Vue, React, AngularJS, Unity, Cocos,...

Databases	MySQL, PostgreSQL, , MongoDB, SQL Server, Redis,...
AI & Cloud	OpenAI API, Gemini API, RAG Architecture, ComfyUI, LLM Integration
Others	Git, Docker, ChromeAPIs, NAS Administration, Kubernetes,...
Languages	English (IELTS 6.5 – CEFR B2) Japanese (Elementary – JPD113 & JPD123)

CERTIFICATIONS

User Experience Research and Design - University of Michigan	2025
Project Management Project - University of California, Irvine	2025
Certified Ethical Emerging Technologist - CertNexus	2024
Software Development Lifecycle - University of Minnesota	2024
Web Design for Everybody: Basics of Web Development & Coding - University of Michigan	2023
Computer Communications - University of Colorado System	2023

ACTIVITIES

Vietnam VEX Robotics National Championship 2024.01 - 2024.01

Volunteer

Responsible for ensuring a smooth and efficient competition flow and maintaining a professional demeanor throughout the event.

Algorithm Arena 4.0 2022.12 - 2023.03

Vice Head Organizer

o Prestigious event that showcases the skills and talents of programmers.
o Organizer of the 4th annual competitive programming contest of JS Club .